ab7501b2-4582-47ce-ad75-562938b45365

<prev_user>

<request>

You are working on an issue in the 'qian-o/Metal.NET' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/Metal.NET/Metal.NET (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-11T13:06:58.203Z</current_datetime>

Consider the following problem statement:

<problem_statement>
I need you to check all the structs completely and compare them with the previous version to see if there is any missing content, and if there are any missing references where the structs are used.
</problem_statement>

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.

<prev_assistant>

The comprehensive struct comparison is complete. Here's a summary of findings and the one fix applied:

Summary

Complete Struct Audit Results

47 total structs across 3 files — all accounted for and verified

against metal-cpp headers:
- **42 auto-generated** in `Metal/MTLStructs.cs` (from metal-cpp `_MTL_PACKED` structs)
- **2 auto-generated** in `Foundation/NSStructs.cs` (from metal-cpp `_NS_PACKED` structs)
- **3 hand-maintained** in `Metal/ExternalStructs.cs` (CGSize, SimdFloat4, SimdFloat4x4 — not in metal-cpp)

Removed structs (correct — don't exist in metal-cpp)
- `MTL4Origin`, `MTL4Range`, `MTL4Size` — these were manual approximations; metal-cpp uses `MTL::Origin`/`NS::Range`/`MTL::Size` directly
- `MTLRange` — no such type in metal-cpp (only `NS::Range`)

Fix Applied
Removed stale `"MTLRange"` entry from `TypeMapper.StructTypes` in the generator — the struct no longer exists and was a leftover from the manual era.

No Missing References
No generated or hand-written code references any of the removed struct types. Build succeeds cleanly.

<current>

<request>

You are working on an issue in the 'qian-o/Metal.NET' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/Metal.NET/Metal.NET (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-11T13:26:28.857Z</current_datetime>

Consider the following problem statement:

<problem_statement>
Great, next optimize all the content you wrote and make its code style more consistent. For the manually written structures, I suggest putting them in the Common directory so that they won't be mixed with the auto-generated code.
</problem_statement>

```
Implement the necessary changes to the repository so that the
requirements specified in the <problem_statement> are met.
```